

Question 1: If G is a graph, prove that $\Gamma(G) = \Gamma(\overline{G})$

Let $\alpha \in \Gamma(G)$ be an automorphism. Then

$$uv \in E(G) \iff \alpha(u)\alpha(v) \in E(G)$$

The negation of this is:

$$uv \notin E(G) \iff \alpha(u)\alpha(v) \notin E(G)$$

But this is the same as:

$$uv \in E(\overline{G}) \iff \alpha(u)\alpha(v) \in E(\overline{G})$$

That is, $\alpha \in \Gamma(\overline{G})$. Since this is an if and only if, the other direction holds and so $\Gamma(G) = \Gamma(\overline{G})$

Question 2: Let G be a graph of order n that is neither complete nor empty. If $G \neq C_4$, prove that $|\Gamma(G)| \leq (n-1)!$

(Empty graph is $\overline{K_n}$, so an edgeless graph)

TODO

Question 3: Prove that *is similar* is an equivalence relation on the vertex set of a graph

We will use $\sim$ to denote *is similar*. Let G be a graph and $v \in V(G)$. Clearly $\text{Id} \in \Gamma(G)$ so $v \sim \text{Id}(v) = v$ (reflexivity). Suppose $v \sim w$. Then there is an automorphism $\alpha \in \Gamma(G)$ such that $\alpha(v) = w$. But $\Gamma(G)$ is a group, so $\alpha^{-1}(w) = v$ and $\alpha^{-1} \in \Gamma(G)$ so $w \sim v$ (symmetry). Now suppose $u \sim v$ and $v \sim w$, corresponding to automorphisms α and β . Then, since $\Gamma(G)$ is a group under composition, $\beta\alpha \in \Gamma(G)$ and $(\beta\alpha)(u) = \beta(\alpha(u)) = \beta(v) = w$ so $u \sim w$ (transitivity).

Question 4: Two edges ab and xy of a graph G are *similar* if there is an automorphism α such that either (i) $a^\alpha = x$ and $b^\alpha = y$, or, (ii) $a^\alpha = y$ and $b^\alpha = x$. If every edge is similar to every other edge, then the graph is called *edge-transitive*

(a) Find a graph that is edge-transitive but not vertex-transitive

$K_{2,3}$

Question 5: N/A

Question 6: N/A (group theory)

Question 7:

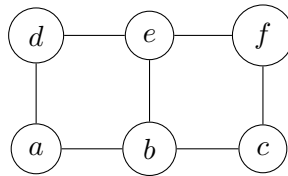
(a) Find a graph that is vertex-transitive but not edge-transitive

Two copies of the complete graph K_n for $n \geq 3$, that are linked by n edges (one edge between v_i and v'_i for each i) is vertex-transitive but not edge-transitive.

Question 8: Let $G = P_3 \times P_2$ and let Γ be the automorphism group of G

(a) Use cycle notation to list the automorphisms in Γ as permutations of $V(G)$.

First notice our graph looks like:



Noticing the symmetries of our graph, we have automorphisms

$$(a), \quad (ad)(be)(cf), \quad (ac)(df), \quad (af)(dc)(be)$$

(b) Determine the orbits

$$a^\Gamma = \{a, d, c, f\}$$

$$b^\Gamma = \{b, e\}$$

$$c^\Gamma = \{a, d, c, f\}$$

$$d^\Gamma = \{a, d, c, f\}$$

$$e^\Gamma = \{b, e\}$$

$$f^\Gamma = \{a, d, c, f\}$$

(c) For each vertex v of G , find Γ_v .

Recall $\Gamma_v = \{\alpha \in \Gamma : v^\alpha = v\}$. So

$$\Gamma_a = \{(a)\}$$

$$\Gamma_b = \{(a), (ac)(df)\}$$

$$\Gamma_c = \{(a)\}$$

$$\Gamma_d = \{(a)\}$$

$$\Gamma_e = \{(a), (ac)(df)\}$$

$$\Gamma_f = \{(a)\}$$

(d) For each automorphism $\alpha \in \Gamma$, determine $\text{fix}(\alpha)$. Hence use the Cauchy-Frobenius-Burnside Theorem to determine the number of orbits.

We have that

$$\text{fix}((a)) = \{a, b, c, d, e, f\}$$

$$\text{fix}((ad)(be)(cf)) = \emptyset$$

$$\text{fix}((ac)(df)) = \{e, b\}$$

$$\text{fix}((af)(dc)(be)) = \emptyset$$

Thus by the Cauchy-Frobenius-Burnside Theorem, the number of orbits is:

$$\frac{1}{|\Gamma|} \sum_{\alpha \in \Gamma} |\text{fix}(\alpha)| = \frac{1}{4}(8) = 2$$

(e) Use the Cauchy-Frobenius-Burnside Theorem to determine the number of distinct 2-colourings of G .

We determine the number of orbits of the automorphism group acting on the set of all 2-colourings. There are a total of $2^6 = 64$ 2-colourings. Now

$$|\text{fix}((a))| = 64$$

$$|\text{fix}((ad)(be)(cf))| = 2 \times 2 \times 2 = 8$$

$$|\text{fix}((ac)(df))| = 2 \times 2 \times (2 \times 2) = 16$$

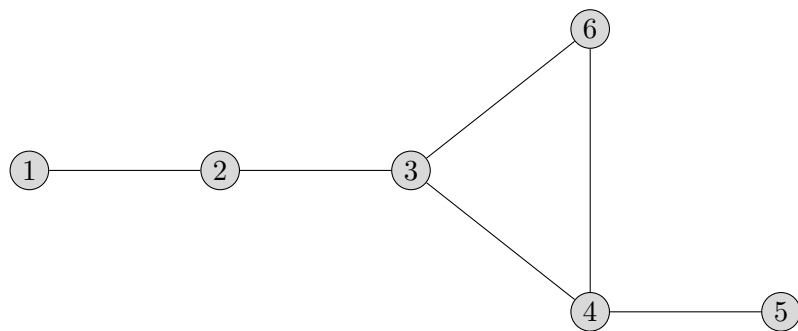
$$|\text{fix}((af)(dc)(be))| = 8$$

So the number of orbits (and thus distinct 2-colourings) is

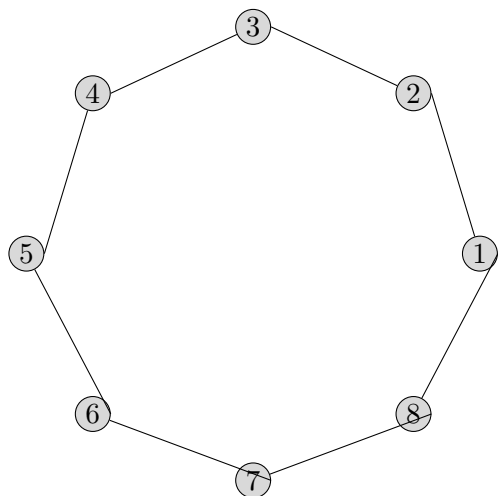
$$\frac{1}{4}(96) = 24$$

Question 9: The tree T in Example 7.1.5 is the smallest nontrivial tree with a trivial automorphism group. Is T the smallest nontrivial graph with a trivial automorphism group?

No it is not.



Question 10: Use the Cauchy-Frobenius-Burnside Theorem to determine the number of distinct 3-colourings of C_8 .



The automorphism group is the dihedral group, of order 16 (1 identity + 7 rotations + 8 reflections). There are 3^8 3-colourings. Each rotation fixes:

$$1 = 3 \text{ colourings}$$

$$2 = 3^2 = 9 \text{ colourings}$$

$$3 = 3 \text{ colourings}$$

$$4 = 3^4 = 81 \text{ colourings}$$

$$5 = 3 \text{ colourings}$$

$$6 = 3^2 = 9 \text{ colourings}$$

$$7 = 3 \text{ colourings}$$

4 reflections fix 3^5 colourings (reflections through vertices) and 4 reflections fix 3^4 colourings (reflections through midpoints). So, by the Cauchy-Frobenius-Burnside theorem, there are:

$$\frac{1}{16}(3^8 + (3 + 9 + 3 + 81 + 3 + 9 + 3) + (4 \times 3^5 + 4 \times 3^4)) = 498$$

Question 11: Use the Cauchy-Frobenius-Burnside Theorem to determine the number of graphs of order 5.